Vivek. B. S

2611006900 34

Cryptology Theory assignment-2

1. Encrypt "CHASE YOUR DREAMS" using

a) Ceasar cipher.

$n \to 3$

$c \to (p+3) \bmod 26$

PT	p+3	(p+3) mod 26	CT
C	5	5	F
H	10	10	K
A	3	3	D
S	21	21	V
E	7	7	H
Y	27	1	R
O	17	17	R
U	23	23	X
R	20	20	U
D	6	6	G
R	20	20	U
E	7	7	H
A	3	3	D
M	15	15	P
S	21	21	Y

Ciphertext: FKDVH BRXU GUHDPY

b) shift cipher with key 7

$c \to (p+7) \bmod 26$

PT → CHASE YOUR DREAMS

p+7 → 9 14 7 25 11 31 21 27 24 10 24 11 7 19 25

(p+7) mod 26 → 9 14 7 25 11 5 21 1 24 10 24 11 7 19 25

Ciphertext:- JOH2L FVBY KYLHTZ

c) Rot 13

PT → C H A S E Y O U R D R E A M S

P+13 → 15 20 13 31 17 37 27 33 30 16 30 17 13 25 31

(P+13) mod26 → 15 20 13 5 17 11 1 7 4 16 4 17 13 25 5

Cipher text :- P U N F R L B H E Q E R N Z F

2. Decrypt

a) "WRWDOOB VHFVHW PH VVDJH" using ceaser cipher.

CT → W R W D O O B V H F V H W P H V V D J H

22 17 22 3 14 14 1 21 7 5 20 7 22 15 7 21 21 3 9 7

P(c-3) → 19 14 19 0 11 11 -2 18 4 2 17 4 19 12 4 18 18 0 6 4

(c-3) mod26 → 19 14 19 0 11 11 24 18 4 2 17 4 19 12 4 18 18 0 6 4

Plain text → T O T A L L Y S E C K E T M E S S A G E

b) "MJQQT BTWQI" shift cipher 5

CT → M J Q Q T B T W Q I

12 9 16 16 19 1 19 22 16 8

C-5 → 7 4 11 11 14 -4 14 17 11 3

(C-5) mod 26 → 7 4 11 11 14 22 14 17 11 3

Plain text : H E L L O W O R L D

" S J B D T W P Y N R J " shift cipher 5

CT →	S	J	B	D	T	W	P	Y	N	R	J
	18	9	1	3	19	22	15	24	13	17	9
L-5 →	13	4	-4	-2	14	17	10	19	8	12	4
(L-5) mod 26 →	13	4	22	24	14	17	10	19	8	12	4

Plain text : NEW YORK TIME

c) " F R A Q 2 B E R P B S S R R " using ROT 13

CT →	F	R	A	Q	2	B	E	R	P	B	S	S	R	R
	5	17	0	16	25	1	4	17	15	1	18	18	17	17
(L-13) →	-8	4	-13	3	12	-12	-9	4	2	-12	5	5	4	4
(L-13) mod 26 →	18	4	13	3	12	14	17	4	2	14	5	5	4	4

Plaint text : S E N D M O R E C O F F E E